Notes on Computational Fluid Dynamics: General Principles

4.2 Fixed value and fixed gradient

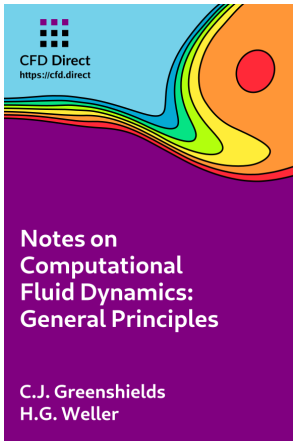
In an equation for $\boldsymbol{\Psi}$ that is discretised to form a matrix equation $[\mathbf{A}][\boldsymbol{\Psi}] = [\mathbf{b}]$, there are two terms that include properties interpolated to faces:

- an advection term of the form $\nabla \bullet (\mathbf{u}\boldsymbol{\Psi})$ which is discretised by Eq. (3.8) in extensive form (see Sec. 3.5) as $\sum_\mathrm{f} \phi_\mathrm{f} \boldsymbol{\Psi}_\mathrm{f}$;
- a Laplacian term of the form $\nabla \bullet (\Gamma \nabla \boldsymbol{\Psi})$ which is discretised by Eq. (3.2) in extensive form as $\sum_\mathrm{f} |\mathbf{S}_\mathrm{f}| \Gamma_\mathrm{f} \nabla_\mathbf{n} \boldsymbol{\Psi}_\mathrm{f}$.

The advection term requires the value $\boldsymbol{\Psi}_\mathrm{f}$ and Laplacian term requires the surface normal gradient $\nabla_\mathbf{n} \boldsymbol{\Psi}_\mathrm{f}$ at faces. When a face is part of a boundary patch, any gradient $\nabla_\mathbf{n} \boldsymbol{\Psi}_\mathrm{f}$ and/or value $\boldsymbol{\Psi}_\mathrm{f}$ must be specified through boundary conditions.

The **fixed value**, or *Dirichlet* condition,[1] is the first type of boundary condition, where the boundary value $\boldsymbol{\Psi}_\mathrm{b}$ is specified. For example, at an inlet patch, we might specify a temperature $T_\mathrm{b} = 300$ K of the fluid flowing into the domain.

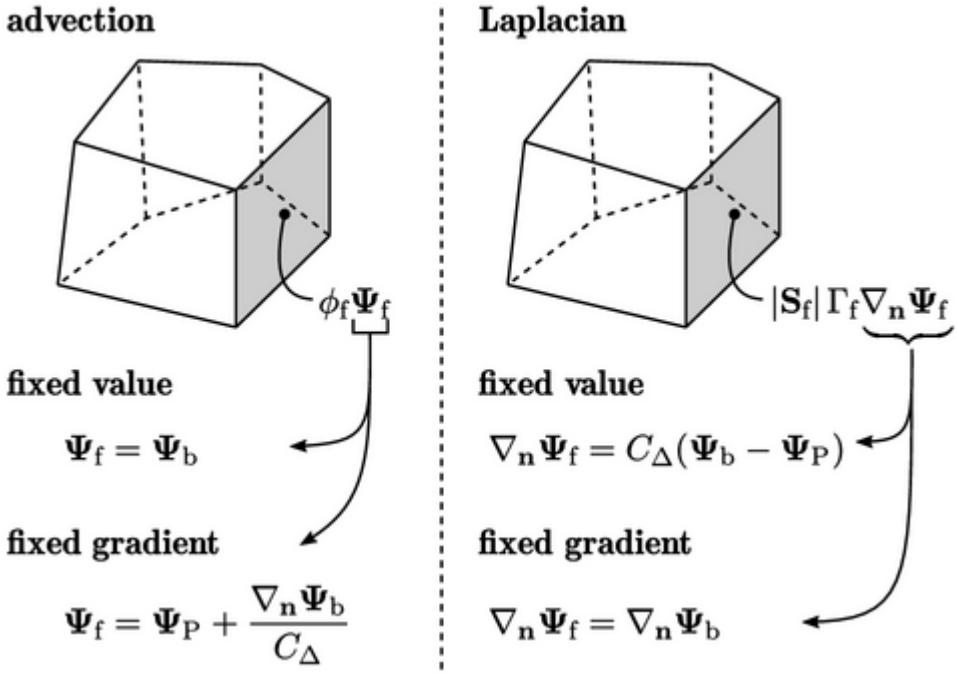
The **fixed gradient**, or *Neumann* condition,[2] is the second type, in which the gradient normal to the boundary $\nabla_\mathbf{n} \boldsymbol{\Psi}_\mathrm{b}$ is specified (where $\nabla_\mathbf{n} \equiv \mathbf{n} \bullet \nabla$). In many cases, the applied normal gradient is zero, which is a common condition applied to many fields, including T, at an outlet patch, *i.e.* where the fluid flows out of the domain.



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When an advection term is discretised, a fixed value condition is applied by substituting the face value Ψ_f with the patch value Ψ_b, *i.e.* setting $\Psi_f = \Psi_b$. When a fixed gradient $\nabla_n \Psi_b$ is specified, the face value is expressed as follows, where Ψ_P is the value in the cell adjacent to each face:

$$\Psi_f = \Psi_P + \frac{\nabla_n \Psi_b}{C_\Delta}. \tag{4.2}$$

When a Laplacian term is discretised, a fixed gradient boundary condition is applied by substituting the face normal gradient $\nabla_n \Psi_f$ with the patch face normal gradient $\nabla_n \Psi_b$, *i.e.* setting $\nabla_n \Psi_f = \nabla_n \Psi_b$. When a fixed value Ψ_b is specified, the face normal gradient is expressed by

$$\nabla_n \Psi_f = C_\Delta(\Psi_b - \Psi_P). \tag{4.3}$$

[1] Peter Dirichlet, *Über einen neuen Ausdruck zur Bestimmung der Dictigkeit einerunendlich dünnen Kugelschale, wenn der Werth des Potentials derselben in jedem Punkte ihrer Oberfläche gegeben ist*, 1850.

[2] after Carl Neumann 1832-1925.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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